

Boundary agreement determines maximal-minor integrals

MathlibAnnex.NullLagrangian.integral_maximalMinor_eq_of_pointwise_boundary_eq

theorem

Turns pointwise equality on a seminorm sphere into a compactly supported Lipschitz perturbation.

Statement

Fix $n \geq 1$, and let p be a continuous real seminorm on $\mathbb{R}^n$ whose closed unit ball $K = \{x : p(x) \leq 1\}$ is compact. Let $F, G : \mathbb{R}^n \rightarrow \mathbb{R}^N$ be globally Lipschitz and satisfy $F(x) = G(x)$ whenever $p(x) = 1$. For a fixed increasing selection $s = \{s_1 < \dots < s_n\}$ of output coordinates, put $\pi_s(y) = (y_{s_1}, \dots, y_{s_n})$ and $M_s(f) = \det(\pi_s \circ Df)$. Then

$$\int_K M_s(F)(x) dx = \int_K M_s(G)(x) dx.$$

Assumptions

There are constants $C_F, C_G \geq 0$ for which

$$\|F(x) - F(y)\|_\infty \leq C_F \|x - y\|_\infty, \quad \|G(x) - G(y)\|_\infty \leq C_G \|x - y\|_\infty \quad (x, y \in \mathbb{R}^n).$$

The other hypotheses are continuity of p , compactness of $\{p \leq 1\}$, and pointwise equality $F(x) = G(x)$ on $\{p = 1\}$. No smoothness or strict convexity of that sphere, Sobolev trace hypothesis, or separate definiteness assumption on p is added.

Conclusion

The integrals of the same signed, increasingly ordered maximal minor agree on K . Each integral is finite by the Lipschitz derivative bound on the compact set.

The sphere has measure zero for a specific reason: it is the frontier of the open convex seminorm ball, which is a neighborhood of the origin. Compactness of a general set would not by itself imply that its boundary has measure zero.

Proof route

Extend the boundary-zero difference by zero, prove its Lipschitz bound across the sphere, and apply the compact-perturbation identity to a patched map.

▼ Proof steps

1. Put $u = F - G$ and define $u_0(x) = u(x)$ on K , $u_0(x) = 0$ outside. The map u is $(C_F + C_G)$ -Lipschitz and vanishes on $\{p = 1\}$. If $x \in K$ and $y \notin K$, continuity of p along the segment from x to y supplies $z = (1 - t)x + ty$, with $0 \leq t \leq 1$, such that $p(z) = 1$. For $p(x) = 1$ take $z = x$. Since $\|x - z\| = t\|x - y\|$, the mixed-side estimate is

$$\|u_0(x) - u_0(y)\| = \|u(x) - u(z)\| \leq (C_F + C_G)\|x - z\| \leq (C_F + C_G)\|x - y\|.$$

The same-side cases follow from the bound for u or from both values being zero; reversing x, y handles the other mixed case. Thus u_0 is globally Lipschitz with this constant. Its topological support satisfies $\text{supp } u_0 := \overline{\{x : u_0(x) \neq 0\}} \subseteq K$, because K is closed. It is therefore compact.

2. Let $H = G + u_0$. Then $H = F$ on K and $H = G$ outside. Apply the Lipschitz compact-perturbation theorem with base G , perturbation u_0 , and the fixed s . The hypotheses just verified give

$$\int_{\mathbb{R}^n} (M_s(H) - M_s(G)) = 0.$$

This difference is integrable: its restriction to the compact support is a difference of integrable Lipschitz minors, and it vanishes outside that support.

3. To compare derivatives, use local equality, rather than merely equality at one point. On the open set $\{p < 1\}$, H agrees locally with F , hence $DH = DF$. On the open complement of K , H agrees locally with G , hence $DH = DG$. The excluded set is $\{p = 1\}$. The seminorm gauge identity identifies it with the frontier of $\{p < 1\}$; this set is convex and contains a neighborhood of 0 by continuity and the positive radius. The convex-frontier Haar-measure theorem therefore gives $|\{p = 1\}| = 0$. These are the hypotheses used by the linked private sphere-null lemma.

4. The local identities from Step 3 give

$$M_s(H) - M_s(G) = M_s(F) - M_s(G) \quad \text{almost everywhere on } K,$$

and $M_s(H) - M_s(G) = 0$ on K^c . The integrable difference from Step 2 can therefore be split as

$$\begin{aligned} 0 &= \int_{\mathbb{R}^n} (M_s(H) - M_s(G)) \\ &= \int_K (M_s(H) - M_s(G)) + \int_{K^c} (M_s(H) - M_s(G)) \\ &= \int_K (M_s(F) - M_s(G)) + 0 \\ &= \int_K M_s(F) - \int_K M_s(G). \end{aligned}$$

The last line uses the integrability of each Lipschitz minor on compact K . This proves the claimed equality.

Main citations

- Exact declaration and its source —
MathlibAnnex.NullLagrangian.integral_maximalMinor_eq_of_pointwise_boundary_eq
- Lipschitz compact-perturbation identity —
MathlibAnnex.NullLagrangian.integral_maximalMinor_fderiv_add_sub_eq_zero_of_lipschitz
With

- Selected minor and its compact-set integrability — `MathlibAnnex.NullLagrangian.integrableOn_maximalMinor_fderiv_of_lipschitzWith`
- Selected maximal-minor integrand — `MathlibAnnex.NullLagrangian.maximalMinorIntegrand`
- Boundary point on a segment (private helper) — `MathlibAnnex.NullLagrangian.exists_boundary_point_between`
- Lipschitz zero extension (private helper) — `MathlibAnnex.NullLagrangian.lipschitz_zeroExtension`
- Null seminorm sphere (private helper) — `MathlibAnnex.NullLagrangian.modelSphereSet_null`
- Patched-map integral argument (private helper) — `MathlibAnnex.NullLagrangian.topMinor_boundary_trace`

Lean source signature (exact)

```
theorem integral_maximalMinor_eq_of_pointwise_boundary_eq
  {m N : ℕ} (p : Seminorm ℝ (Fin (m + 1) → ℝ)) (hp : Continuous p)
  (hK : IsCompact (p.closedBall 0 1))
  (s : Matrix.MaximalMinorIndex (m + 1) (Fin N))
  {F G : (Fin (m + 1) → ℝ) → (Fin N → ℝ)} {CF CG : ℝ≥0}
  (hF : LipschitzWith CF F) (hG : LipschitzWith CG G)
  (htrace : ∀ x, p x = 1 → F x = G x) :
  ∫ x in p.closedBall 0 1, maximalMinorIntegrand s F x =
  ∫ x in p.closedBall 0 1, maximalMinorIntegrand s G x
```

In the source	Mathematical meaning
<code>{m N : ℕ} (p : Seminorm ℝ (Fin (m + 1) → ℝ)) (hp : Continuous p)</code>	The positive dimension $n = m + 1$ and continuous seminorm p on $\mathbb{R}^n$.
<code>(hK : IsCompact (p.closedBall 0 1))</code>	The set $K = \{x : p(x) \leq 1\}$ is compact; no separate definiteness hypothesis is added.
<code>(s : Matrix.MaximalMinorIndex (m + 1) (Fin N))</code>	The fixed increasing selection s of n output coordinates, with π_s and signed minor $M_s(f) = \det(\pi_s \circ Df)$ as defined in this Card.
<code>{F G : (Fin (m + 1) → ℝ) → (Fin N → ℝ)} {CF CG : ℝ≥0} (hF : LipschitzWith CF F) (hG : LipschitzWith CG G)</code>	The maps F, G have the stated global sup-norm Lipschitz bounds with constants $C_F, C_G \geq 0$.
In the source	Mathematical meaning
<code>(htrace : ∀ x, p x = 1 → F x = G x)</code>	Pointwise equality $F(x) = G(x)$ at every x on the seminorm sphere $\{p = 1\}$. This is neither an a.e. condition nor a Sobolev trace.

In the source	Mathematical meaning
$\int x$ in $p.\text{closedBall } 0 \ 1, \text{ maximalMinorIntegrand}$ $s \ F \ x = \int x$ in $p.\text{closedBall } 0 \ 1, \text{ maximalMinorIntegrand } s \ G \ x$	The conclusion $\int_K M_s(F) dx = \int_K M_s(G) dx$: same compact domain, Lebesgue measure, increasing selection and signed minor on both sides.

▼ Exact surrounding binder context (separate excerpt)

```

noncomputable section
open Set Function MeasureTheory Filter
open scoped BigOperators Topology NNReal
namespace MathlibAnnex
namespace NullLagrangian

private structure SeminormBall (n : ℕ) where
  p : Seminorm ℝ (Fin n → ℝ)
  continuous_p : Continuous p
  isCompact_closedBall : IsCompact (p.closedBall 0 1)

namespace SeminormBall

```

▼ Exact content identity

Declaration: `MathlibAnnex.NullLagrangian.integral_maximalMinor_eq_of_pointwise_boundary_eq`

Accepted content SHA-256: 0149c1ed2c440df531b6391bff4bc0fdcadb76b36f1764e2fde5fa3a8cabb6e7

Accepted source guide SHA-256: 526fb4ea936abf392ef576e64ad13117584ddb161fe88cac0343d4032842a630

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73